PRACTICAL COURSE WORKBOOK

SQL for Financial Analysis

Model, query and explain dependable data

LEVEL	Intermediate
GUIDED TIME	5 hours across 12 lessons
PROJECT	a working data model with validated queries
SKILLS	SQL, Finance, Analytics, Window functions, Reporting, Critical evaluation
EDITORIAL STATUS	Structured draft v0.9

WHAT YOU WILL BE ABLE TO DO

- Use SQL appropriately in a realistic, bounded task.
- Use Finance appropriately in a realistic, bounded task.
- Use Analytics appropriately in a realistic, bounded task.
- Use Window functions appropriately in a realistic, bounded task.

WHO THIS IS FOR

Developers and analysts who need dependable data systems.

BEFORE YOU BEGIN

- Basic programming or spreadsheet experience

This open workbook is a structured draft undergoing course-specific editorial review. It does not provide certification.

COURSE MAP

From foundation to finished work

SQL for Financial Analysis is a structured, practical course covering SQL, Finance, Analytics, Window functions, Reporting. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

MODULE	LESSONS AND FOCUS
01 SQL: Data modelling	1. Entities with SQL 2. Relationships with Finance 3. Constraints with Analytics
02 Finance: Querying	1. Selection with Window functions 2. Joins with Reporting 3. Aggregation with SQL
03 Analytics: Reliability	1. Transactions with Finance 2. Indexes with Analytics 3. Backups with Window functions
04 Window functions: Production design	1. Security with Reporting 2. Performance with SQL 3. Operations with Finance

HOW TO STUDY EACH LESSON

1. Read the objective and identify the decision it supports.
2. Reproduce the worked example with the stated input.
3. Test one normal case and one boundary or failure case.
4. Complete the knowledge check without guessing.
5. Save a short evidence note: result, limitation and next action.

CAPSTONE PROJECT

Produce a reviewable SQL for Financial Analysis project using SQL, Finance, Analytics.

DELIVERABLE	A working SQL for Financial Analysis artefact plus an evidence-based self-review.
TOOLS	A suitable SQL environment, A plain-text decision log, Test data or realistic sample material

Production plan

1. Define the intended user, outcome and constraints.
2. Create the smallest complete result using SQL.
3. Test one normal case, one boundary case and one failure response.
4. Revise the work from the evidence and preserve before-and-after results.
5. Prepare a concise handover containing method, limitations and next step.

Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

Evidence log

INPUT	Original brief, data, requirements or test material.
DECISIONS	Chosen method and one reasonable alternative.
TESTS	Normal, boundary and failure results.
LIMITATION	What the result does not establish.
REVISION	One evidence-led improvement.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

Authoritative references

PostgreSQL Tutorial

PostgreSQL Global Development Group - reviewed 2026-08-21
https://www.postgresql.org/docs/current/tutorial.html

SQL Language

PostgreSQL Global Development Group - reviewed 2026-08-21
https://www.postgresql.org/docs/current/sql.html

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the SQL scope.